Shayla Mumford

Seattle, WA | (206)930-2719 | mumfords@spu.edu | linkedin

Qualifications

- Completed high-level coursework: Statics, Dynamics, Mechanics of Materials, Thermodynamics, Fluid Mechanics, Heat Transfer, etc.
- Designed, fabricated, and tested Baja SAE off-road vehicle components under dynamic loading and durability requirements
- Developed 3D CAD models and assemblies, generated engineering drawings for manufacturing
- Performed component testing, troubleshooting, and design iteration of multiple manufacturing jigs to aid in safety procedures

Education

Bachelor of Science in Mechanical Engineering

Minor: Appropriate and Sustainable Engineering

Seattle Pacific University | GPA 3.87 | Dean's List: September 2023 – Present *Exp. Graduation June 2027*

Leadership & Design Experience

Baja SAE SPU Falcon Racing – Front Drivetrain Lead | Seattle Pacific University

June 2024 – Present

- Led a project team of 5-10 students redesigning the front end of the drivetrain on SPU Falcon Racing's Baja SAE vehicle to allow for 4WD capabilities
- Cross-communicated with other leads for smooth vehicle integration

Engineering Projects

- Junior design project: Designed and manufactured a temperature- and humidity-controlled testing chamber to assist electronic developers
- Designed and 3D printed a full magnet-installation jig to assist in safe manufacturing processes

Work Experience

Manufacturing Engineer Intern | Vista Clara Inc. – Mukilteo, WA

June 2026 - September 2026

- Manufactured different NMR borehole logging units, which included surface control units as well as downhole logging sensors
- Revised assembly instructions and pre-existing SolidWorks models and assemblies for clarity and accuracy
- Created multiple comprehensive SolidWorks assemblies for full sensor probes and surface units, which I then used for both BOM and 2D drawing generation.

Sales Associate – Children's Apparel | Nordstrom - Bellevue, WA

July 2025 - June 2026

- Worked with team members in a fast-paced environment to support efficient operations.
- Evaluated product details and specifications to provide accurate recommendations.
- Input and maintained precise inventory data in a database system.

Field Station Student Worker | Seattle Pacific University - Blakely Island, WA

June 2024 - October 2024

- Performed preventive maintenance and repairs on field station vehicles, including oil changes, belt and fan replacements, and system inspections to ensure safe operation.
- Assisted with mechanical troubleshooting of small engines, HVAC units, and facility systems
- Supported day-to-day facility upkeep (cooking, cleaning, equipment prep), contributing to a safe and functional living/working environment for staff and students.

Relevant Skills

- 3D CAD & Modeling: SolidWorks and OnShape
- Simulation & Analysis: MATLAB, Octave, Simulink, SolidWorks Simulation (FEA), Arduino, and Algadoo
- Prototyping & Fabrication: 3D printing, machining, welding, assembly
- Testing & Validation: Mechanical system testing, data collection, and performance evaluation in Excel
- Teamwork & Leadership: Cross-functional collaboration, technical documentation, project leadership